

Game Theory – 2

Strategy stealing, pairing and invariants; games with a value

14 сентября 2026 года

Theoretical minimum. The Sprague–Grundy theory of Game Theory – 1 covers impartial finite games under the normal convention and nothing else. Everything outside that class is won by one of four arguments, and this sheet is a catalogue of them.

Theorem 1 (strategy stealing). Suppose a game is symmetric in the two players, has no draws, and has the monotonicity property that owning an extra move is never a disadvantage. Then the first player has a winning strategy. *Proof:* if the second player had a winning strategy, the first player could make an arbitrary first move and then follow it, treating his own extra mark as a bonus; he would then win as well, which is impossible. *The argument is purely non-constructive:* for Chomp it gives no winning move for any board larger than a rectangle strip, and this is the standard example of a proof of existence with no algorithm behind it (Methods of proof – 5).

Theorem 2 (pairing and symmetry). If the positions can be split into disjoint pairs so that the second player can always answer inside the pair just used by the first, then the second player at least draws. The mirror strategy on a symmetric board and the “answer in the same row” strategy are the two usual realisations. Pairing arguments prove impossibility results too: if no pairing exists, the count of positions is the obstruction.

Fact 1 (invariants and monovariants). A quantity preserved by every move separates reachable from unreachable positions and so decides the outcome; a quantity that strictly decreases and is bounded below proves that the game terminates, which is exactly the hypothesis of backward induction. Games in which termination is the whole question are decided by a monovariant (Methods of proof – 4).

Theorem 3 (the minimax theorem). A finite two-person zero-sum game

is given by a payoff matrix $A \in \mathbb{R}^{m \times n}$: the first player chooses a row, the second a column, the payoff is a_{ij} . Allowing mixed strategies, that is probability vectors x and y ,

$$\max_x \min_y x^\top A y = \min_y \max_x x^\top A y,$$

and the common value is the *value* of the game. The proof is a separation theorem for convex sets applied to the convex hulls of the rows and the columns (Convex – 3), or equivalently linear-programming duality (Optimization – 1). Pure strategies need not suffice, and the size of the support of an optimal mixed strategy is what the exercises below ask for.

Theorem 4 (infinite games). In a game of infinite length in which the players alternately choose elements of a set and the first player wins if the resulting sequence lands in a set A , determinacy is no longer automatic. Gale and Stewart proved that if A is open (in the product topology, that is, if a win for the first player is always decided after finitely many moves) then the game is determined, by a transfinite version of backward induction; and using the axiom of choice one constructs a set A for which neither player has a winning strategy. (Topology of the real line – 3.)

1. Prove the strategy-stealing theorem and apply it to Chomp: the first player wins on every rectangular board with more than one square. Then prove that a game of Hex can never end in a draw, deduce that the first player wins, and explain precisely why neither argument produces a winning move.
2. Consider a polyhedron with at least five faces in which exactly three edges meet at every vertex. Two players in turn sign their name on a previously unsigned face; the winner is the one who first signs three faces sharing a common vertex. Show that the first player wins with best play. (Putnam 2002)
3. Consider a polynomial $f(x) = x^{2012} + a_{2011}x^{2011} + \cdots + a_1x + a_0$.

Two players in turn choose one of the coefficients $a_0, \dots, a_{2011}$ that has not yet been fixed and assign it a real value; the first player wants to prevent f from being divisible by a fixed polynomial $m(x)$, the second wants to force divisibility. The first player moves first. Who wins for $m(x) = x - 2012$, and who wins for $m(x) = x^2 + 1$? (IMC 2012)

4. Prove the minimax theorem for 2×2 matrices by direct computation, state the general theorem, and find the value and the optimal mixed strategies of the game with payoff matrix $\begin{pmatrix} 0 & 1 & -1 \\ -1 & 0 & 1 \\ 1 & -1 & 0 \end{pmatrix}$. Then show that a game has a saddle point in pure strategies if and only if $\max_i \min_j a_{ij} = \min_j \max_i a_{ij}$.
5. A sphere carries finitely many marked points. Alice and Bob move alternately, Alice first: Alice counts the marked points, say n of them, and marks $n + 1$ further points; Bob then chooses a hemisphere and erases every marked point lying in it, boundary included. Can Bob always guarantee that after finitely many moves no marked points remain? Find the monovariant. (A related infinite game — covering a set $X \subseteq \mathbb{R}$ by segments whose lengths the opponent dictates — is Topology of the real line – 3, homework 2.) (ICMC)